

PROJECT CHOWKI

*Continuous Health Observation
and Water-Kitchen Intelligence*

The campus outbreak radar that detects silent GI epidemics
in **under 6 hours** — not 2 days.



Residential hostels are epidemiological petri dishes.

Between **500 and 5,000 students** share communal dining halls, unmonitored RO water sumps and high-moisture foods – so outbreaks burn through floors while the institution is still reading nothing at all.

48–72 h of student productivity destroyed before institutional recognition

15–20 severe clinic cases accumulate before an outbreak is even named

~85% of cases never reach the clinic — they self-medicate in their rooms

THE CAMPUS HEALTH ICEBERG

SHARE OF TOTAL MORBIDITY → BAR WIDTH

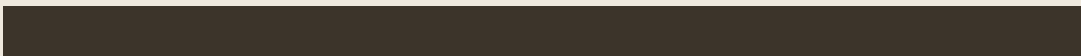
TIER 1 Clinic cases **10–15%**



High fever, severe dehydration – visible only after a 48–72 h lag.

▼ INSTITUTIONAL VISIBILITY LINE – EVERYTHING BELOW IS UNSEEN

TIER 2 Self-medicated **60–75%**



Pharmacy ORS and Ondansetron – bypasses official clinic logging entirely.

TIER 3 Unreported mild **15–25%**



Room rest, peer remedies – zero institutional tracking whatsoever.

WATER FAILURE VECTORS

Biofilm accumulation in overhead tanks · back-siphonage at cross-connections · RO filter bypass under peak demand

FOOD HAZARD VECTORS

Ambient holding of cottage cheese (Palak Paneer) harbouring *Staphylococcus aureus* · raw chutneys harbouring *Salmonella* / STEC

The digital watchpost.

CHOWKI moves campus health from **passive reaction** – waiting for clinic overload – to **active intelligence**, fusing low-friction self-reports, pharmacy point-of-sale surges and inline IoT water quality into one continuous signal.

DIMENSION	STATUS QUO · PASSIVE CLINIC MODEL	PROJECT CHOWKI · ACTIVE AI RADAR
Detection trigger	15–20 severe clinic visits over 2–3 days	3–5 self-reports clustered on one floor or wing
Time to detection	48–72 hours, after community spread	Under 6 hours, at index cluster formation
Methodology	Manual line-listing at the health centre	STPSS scan statistics + Bayesian fusion
Disambiguation	None — cannot separate outbreak from noise	Dual-engine: STPSS gate → Bayesian attribution
Data scope	10–15% of cases — the visible tip only	All three tiers — reports, POS sales, IoT telemetry
Student friction	High — 45-min clinic visit, fear of quarantine	Zero install — 15-second WhatsApp check-in

<6 h

MEAN TIME TO DETECTION
down from 48–72 hours

3–5

CASE ALERT THRESHOLD
down from 15–20 clinic cases

15 s

STUDENT CHECK-IN
zero-install, on WhatsApp

Two gates, no false alarms.

“How do we separate 4 students sick from exam stress from an active *S. aureus* outbreak?”

GATE 1 • SPATIAL TIGHTNESS

Space-Time Permutation Scan Statistics

Scans dynamic space-time cylinders $A = (z, t)$ – no historical population baseline required.

$$LLR_A = c_A \ln(c_A/\mu_A) + (C-c_A) \ln((C-c_A)/(C-\mu_A))$$
$$\mu_A = (C_z \cdot C_t) / C$$

Monte Carlo $N = 999 \rightarrow$ PASS IF $p < 0.05$ AND $LLR > 10$

GATE 2 • CAUSAL ATTRIBUTION

Multi-Parametric Bayesian Attribution

Δt incubation delta vs. meal timestamp
 S symptom vector cosine similarity
 M_k shared mess / water exposure fraction
 E auxiliary sensor prior boost

$$P(O \mid \Delta t, S, M_k, E) \propto P(\Delta t|O) \cdot P(S|O) \cdot P(M_k|O) \cdot P(E|O) \cdot P(O)$$

DECISION RULE $\rightarrow P(\text{Outbreak}) \geq 0.70$

DIAGNOSTIC DIMENSION	GENUINE INFECTIOUS OUTBREAK	COINCIDENTAL UPSETS
STPSS spatial scan	Tight density, $p < 0.05$, $LLR > 10$ — one floor or wing	Scattered across campus, $p \geq 0.05$
Temporal alignment Δt	Compressed 2–8 h or 8–24 h, locked to a mess meal	Dispersed 24–48 h, no meal correlation
Symptom vector S	Co-occurring cluster — vomiting + cramps + fever	Isolated single symptom — mild acidity, heartburn
Exposure hazard M_k	Shared source $M_k > 0.75$ — one dish or sump node	Heterogeneous — off-campus snacks, personal food
Auxiliary telemetry	ORS sales spike $> 200\%$ or chlorine drop < 0.2 mg/L	No POS surge, water chlorine nominal
Final classification	PASSES BOTH GATES $\rightarrow$ ALERT DISPATCHED	FAILS A GATE $\rightarrow$ LOGGED AS BACKGROUND NOISE

Five feeds. All three tiers.

Two of the five channels need *no student action at all* – CHOWKI still sees the outbreak.

01

WhatsApp Micro-Bot

META CLOUD API

15-second conversational check-in. Zero install. Captures symptoms, onset time, meal location and water source.

→ TIER 2 + 3 HIDDEN CASES

02

Student PWA

REACT 18 · WORKBOX

Offline IndexedDB queue engineered for hostel concrete dead zones. Floor advisories and consent controls.

→ TIER 2 + 3 HIDDEN CASES

03

Pharmacy POS API

REST WEBHOOK

Spike alert when ORS / Ondansetron sales exceed **200%** of the 7-day rolling baseline within 6 hours.

→ NO STUDENT ACTION NEEDED

04

Water IoT Sensors

MQTT · 15-MIN INTERVAL

Inline RO sump sensors stream residual chlorine, turbidity, pH and flow. Chlorine < **0.2 mg/L** boosts the Bayesian prior.

→ NO STUDENT ACTION NEEDED

05

Mess Menu Log

WARDEN PORTAL

Daily dishes, serving times and high-risk tags – paneer, raw chutney, cold dairy.

→ FEEDS t_{meal} AND M_k

FUSION PIPELINE



THE 15-SECOND CHECK-IN, VERBATIM

STUDENT “chowki” · BOT Select symptoms → “1 2 4” · BOT Onset window → “B” (2–8 h) · BOT Last meal → “2” (Mess 2) · BOT Water source → “A” (Floor R0)

BOT “Logged anonymously. Current Block B, Floor 3 status: YELLOW ADVISORY.”

REPORTS · 24 H

47

ACTIVE CLUSTERS

3

PRIMARY RISK ZONE

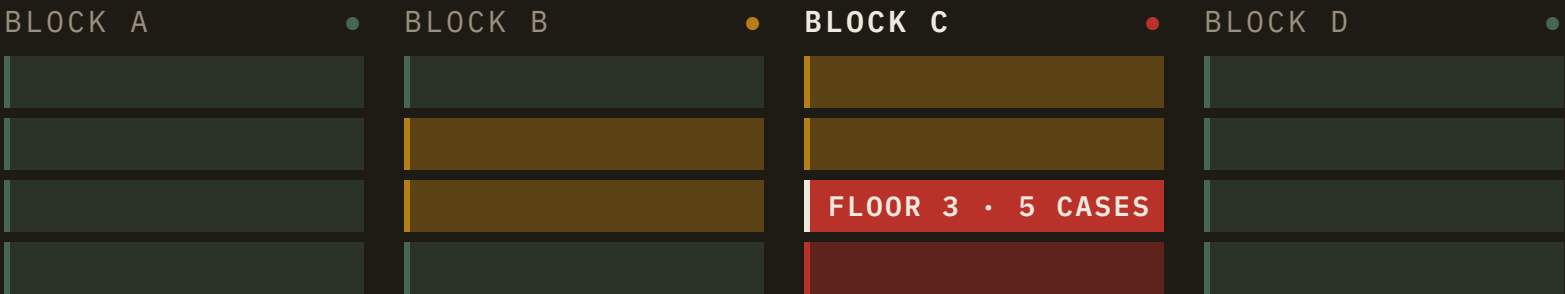
BLOCK C · W2

ALERT STATUS

● LEVEL 2 CONTAINMENT

01 · CAMPUS SPATIAL HEATMAP

FLOOR-LEVEL k-ANONYMOUS BINS



▶ DRILL-DOWN HOSTEL_C_FL_3

LLR 12.4 p < 0.01 RR 6.8× M_k 0.83 k = 7 ✓

02 · EPIDEMIC CURVE

ROLLING 72 H · 4-HOUR BINS · POINT-SOURCE SIGNATURE

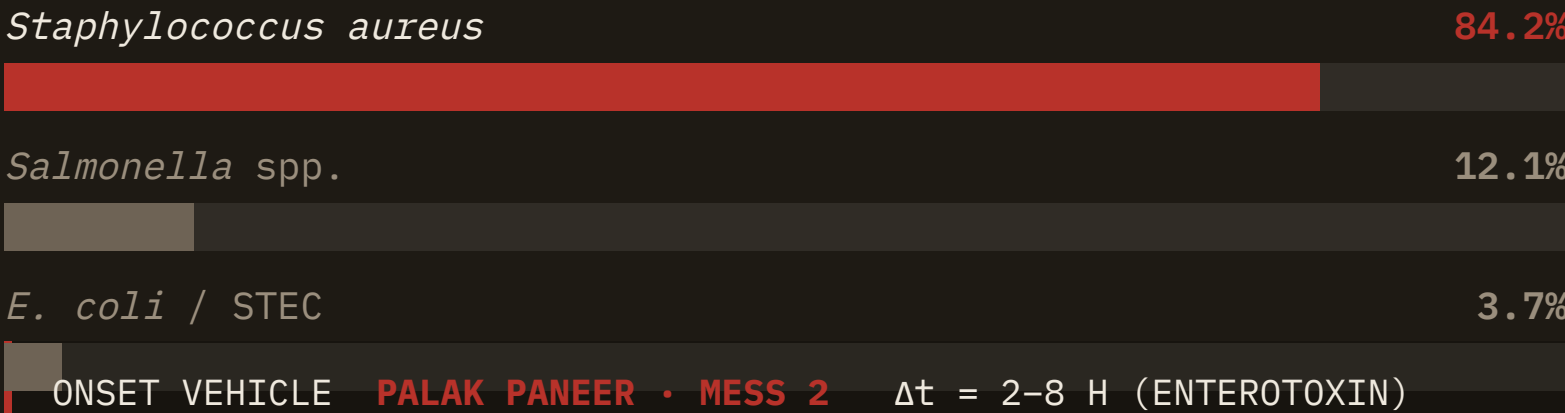
▼ MESS 2 DINNER 19:30 IST
PEAK AT Δt = 4.5 H



FOODBORNE POINT SOURCE, NOT CONTINUOUS EXPOSURE

03 · BAYESIAN PATHOGEN PROFILER

RANKED POSTERIOR PROBABILITY



04 · AUTOMATED SOP DISPATCH

ROLE-GATED · PUSHED ON LEVEL 2 TRIGGER

CMO	Lock Block C RO sump intake	DISPATCHED
MESS	Halt paneer service, Mess 2	DISPATCHED
WARDEN	ORS sachets to Floor 3 desk	IN PROGRESS
STUDENTS	WhatsApp advisory, Block C	214 SENT
SANITATION	Deep-clean Wing 2 washrooms	QUEUED

Surveillance without exposure.

CHOWKI never needs to know *who* is sick — only that a floor-level bin has crossed threshold.

DPDP ACT 2023 · COMPLIANCE MATRIX

PROVISION	IMPLEMENTATION
§ 6(1) Valid consent	Multilingual PWA onboarding with granular affirmative opt-in toggles before any health data is collected
§ 6(5) Right to withdraw	In-app revocation engine triggering an automated 72-hour database purge
§ 7(d) Legitimate use	De-anonymisation restricted to Level 3 emergencies, requiring CMO <i>and</i> DPO co-authorisation
§ 8(5) Security safeguards	AES-256-GCM at rest, TLS 1.3 in transit, pseudonymisation at the client edge
k ≥ 5 Anonymity floor	Minimum 5 anonymous tokens per spatial bin before STPSS will process it — blocks room-level re-identification
TOKENS Identity decoupling	Names and roll numbers stripped at edge, replaced by salted SHA-256 tokens rotating weekly

PRIVACY-BY-DESIGN · FOUR GATES

GATE 1 · EDGE PSEUDONYMISATION

Device strips name, roll number and phone *before transmission*.

```
HMAC-SHA256(salt + roll_no + week_no)
```

GATE 2 · k-ANONYMITY SPATIAL BINNING

Room coordinates collapse into floor bins. Below k = 5, the bin is withheld from the engine entirely.

```
Room 304C → HOSTEL_C_FL_3
```

GATE 3 · RETENTION & PURGE

Pseudonymous records held in PostGIS under an automated TTL schedule.

```
RAW 30 DAYS · AGGREGATE STATS 2 YEARS
```

GATE 4 · EMERGENCY DE-ANONYMISATION

Level 3 only. Dual authorisation by CMO and Data Protection Officer, under § 7(d).

```
EVERY UNLOCK WRITES AN IMMUTABLE AUDIT ENTRY
```

Wardens see **block-level aggregates**. The CMO sees **campus clusters**. Nobody sees a named student — until statute permits it.

Built in 48 hours. Scales to 1,000 campuses.

MEASURABLE IMPACT TARGETS

METRIC	BASELINE	CHOWKI TARGET
Mean time to detection	48-72 h	< 6 h
Cluster detection threshold	15-20 cases	3-5 reports
Secondary attack rate	Uncontrolled	-40 to 60%
Reporting participation	< 15%	> 40%
Check-in completion time	~45 min	< 15 s

HACKATHON SPRINT • HOURS 0-48

H 0-2	Monorepo + Docker Compose: PostGIS, Redis, FastAPI
H 2-8	WhatsApp webhook, student PWA, mess menu API
H 8-16	STPSS scan engine + Bayesian attribution model
H 16-24	CMO command radar, warden SOP panel, mess portal
H 24-48	Data seeder, load test at 50 concurrent users, pitch hardening

POST-HACKATHON • MONTHS 1-6

M 1-2	Clinic line-listing API; Norovirus seasonal engine
M 3-4	Multi-campus federation; IndicBERT multilingual NLP
M 5-6	Kitchen CCTV AI (ambient temperature); IDSP integration
REG	Data Protection Board of India certification; ISO 27001

Every Indian campus already has a चौकी at the gate.
We built the digital one that watches over student health.

TURNKEY DEPLOYMENT

1,000+

RESIDENTIAL CAMPUSES & PG CLUSTERS

Ask us the hard ones.

Q1 · ADOPTION RISK

“What if students simply don’t log their symptoms?”

CHOWKI never depended on self-reporting alone. Two channels are fully passive and fire with **zero student check-ins**:

- ▶ PHARMACY ORS / ONDANSETRON SALES > **200%** BASELINE
- ▶ RO SUMP FREE CHLORINE < **0.2 mg/L** OR TURBIDITY SPIKE

Either alone raises the Bayesian prior and opens an advisory.

Q2 · FALSE POSITIVES

“How do you avoid false alarms from exam-stress belly?”

Stress cases are **spatially dispersed** — they die at Gate 1. If they do cluster, isolated single symptoms die at Gate 2.

- ▶ GATE 1 FAILS $p \geq 0.05$ · NO SPATIAL TIGHTNESS
- ▶ GATE 2 FAILS LOW **S** SIMILARITY · LOW **M_k**

Both gates must pass. One veto is enough to log it as noise.

Q3 · LEGALITY

“Is storing student health data legal under Indian law?”

Yes — DPDP Act 2023 compliant by construction, not by policy document.

- ▶ k-ANONYMITY $k \geq 5$ · EDGE PSEUDONYMISATION
- ▶ WEEKLY ROTATING SALTED SHA-256 TOKENS
- ▶ RAW PURGE **30 DAYS** · WITHDRAWAL ERASURE **72 H**

Q4 · SCALE

“Can this scale to a university of 20,000 students?”

The heavy path is a bounded scan, not a per-student model.

- ▶ STPSS SCAN < **30 s** ACROSS 50,000 SPACE-TIME BINS
- ▶ REDIS 7.2 SUB-MILLISECOND BIN INCREMENTS
- ▶ FASTAPI + POSTGIS 3.4 · HORIZONTALLY SHARDED